pfi

Python for Fixed Income

COURSE 10 · IMPLEMENTING RAG

Retrieval Evaluation: Recall and MRR

WEEK 3 · DAY 1

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Today: scoring retrieval on its own

A labelled set, no model for answers

Chunks	1,158 at 512/64
Questions	25 development, 20 test
Scored	23 and 18 (a page in the key)
Pipelines	MiniLM, BM25, hybrid, + reranker
Requests	none; no key

Today

1. Score retrieval before answers
2. The labelled set
3. evaluate, recall at k, MRR
4. By item: where retrieval misses
5. A miss counted as zero
6. The test questions, once, with an interval
7. What recall does not say

Name the rule of every count; choose nothing today; open the test rows once.

How it works: two families of measures

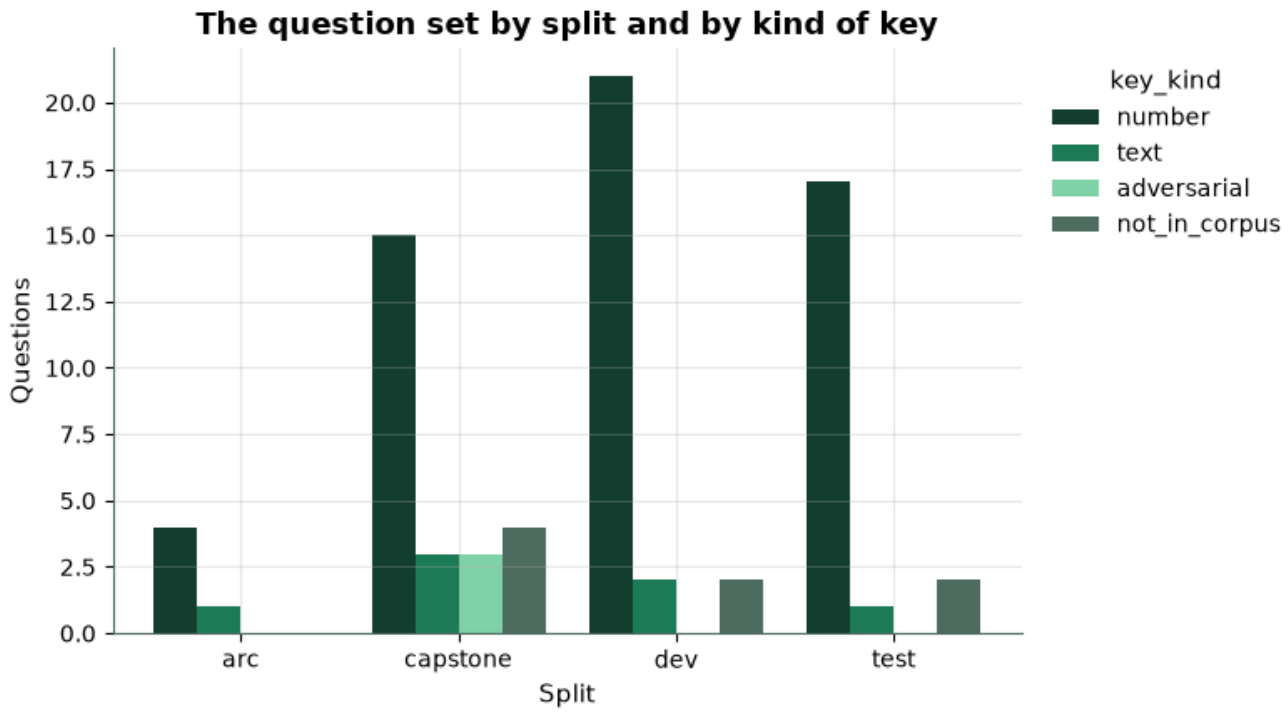
Two failures	The right text never reached the model, or it reached the model and was misread; one answer score cannot say which.
Retrieval	Recall at k: is the key's place among the first k? MRR: how early? Labels of a place, no model, no request.
Answer	Day 1's key grader, day 10's citation check, day 12's judges beside human labels: they read the text a model wrote.
You control	The rule a place is matched by (page or item), k, and which split is read.
What comes out	Two numbers per question that can disagree; the disagreement is the diagnosis.

The two families, in a table

Measure	Family	Reads	Needs	Catches	Cannot see
recall at k	retrieval	the places of the first k chunks against the key's place	a key that names a place; no model	the key's page never reached the model	whether the answer is right
MRR	retrieval	1 / the first place that matches the key's	the same labels; no model	the key's page came late in the list	what the model did with the chunks
key grader (day 1)	answer	the key's figures or text inside the answer	an answer and a key	a wrong figure, a missing figure	a right figure in other units
citation check (day 10)	answer	every sentence's citation against the chunks sent	an answer and the chunks sent	an uncited sentence, a chunk never sent	a cited chunk misread
judges (day 12)	answer	a second model's verdict on grounding and relevance	an answer, the chunks, a request	a claim the chunks do not state	its own limits: day 12 sets it beside human labels

2 retrieval measures run today with no model; 3 answer measures need an answer first. Neither family replaces the other.

Written from pages, never from chunks



One row holds

id, split, question, answer_key

key_kind: number, text, not_in_corpus, adversarial

expected_item, expected_page

key_quote and written_from (a page)

notes

arc 5, capstone 25, dev 25, test 20. 8 of 25 development rows name another place for the figure in their notes (dev01, dev02, dev04, dev09, dev15, dev19, dev20, dev22): the key has one page.

How it works: recall at k and MRR

first_rank	evaluate reads the place of each of the first k chunks in order and records the first that equals the key's, or 0.
recall_at_k	The share of questions with first_rank between 1 and k; a miss counts 0 in the denominator, never dropped.
mrr	The mean of 1 / first_rank, 0 for a miss: place 1 earns 1, place 2 earns 0.5, place 10 earns 0.1. It rewards early.
You control	k, read once at 10 so recall at 1, 3, and 5 come from one run; the rule, by="page" or by="item".
What comes out	One table per pipeline, a row per question, that both functions read and Excel repeats with MATCH and IFERROR.

Every setting of today's functions

Function	Setting	Default	What it controls	Today passes
evaluate	questions	required	the question rows; a row with no place under the rule is left out	dev; test once in 11.6
evaluate	retrieve	required	a retriever: a function from a question to a frame of chunks with doc, page, item	each of the four pipelines, wrapped in remembered
evaluate	k	required	how many places are read; first_rank is 0 past them	10, so recall at 1, 3, and 5 are read from one run
evaluate	by	'page'	the rule a place is matched by: the key's page or its item	"page" (the default); "item" in 11.4
recall_at_k	results	required	the frame evaluate returned, one row per question	one pipeline's results
recall_at_k	k	required	the place a hit must be at or before; a miss stays in the denominator	1, 3, 5, 10
mrr	results	required	the frame evaluate returned; reciprocal_rank is 0 for a miss	one pipeline's results
bootstrap_interval	y_true	required	the outcomes of the scored rows, 0 or 1; a resample with one class only is drawn again	hit at 3 as 0 or 1
bootstrap_interval	proba	required	the values the metric reads, one per row	the hits, or the reciprocal ranks
bootstrap_interval	metric	required	the function metric(y_true, proba) computed on each resample	mean_of: the mean of the values
bootstrap_interval	n_boot	1000	how many resamples	1000
bootstrap_interval	level	0.95	the interval's coverage: the percentiles read are (1 - level) / 2 and (1 + level) / 2	0.95
bootstrap_interval	seed	42	the resampling seed, so the interval repeats	SEED (42)

Three questions by hand, then the functions

id	Key page	Pages of the first 5 places	First place	Hit at 3	Reciprocal rank
dev13	35	35 59 93 93 59	1	yes	1/1 = 1.0000
dev07	38	76 101 76 76 38	5	no	1/5 = 0.2000
dev01	37	40 78 36 19 74	-	no	0 (not found)

The two sums, by hand and by the functions

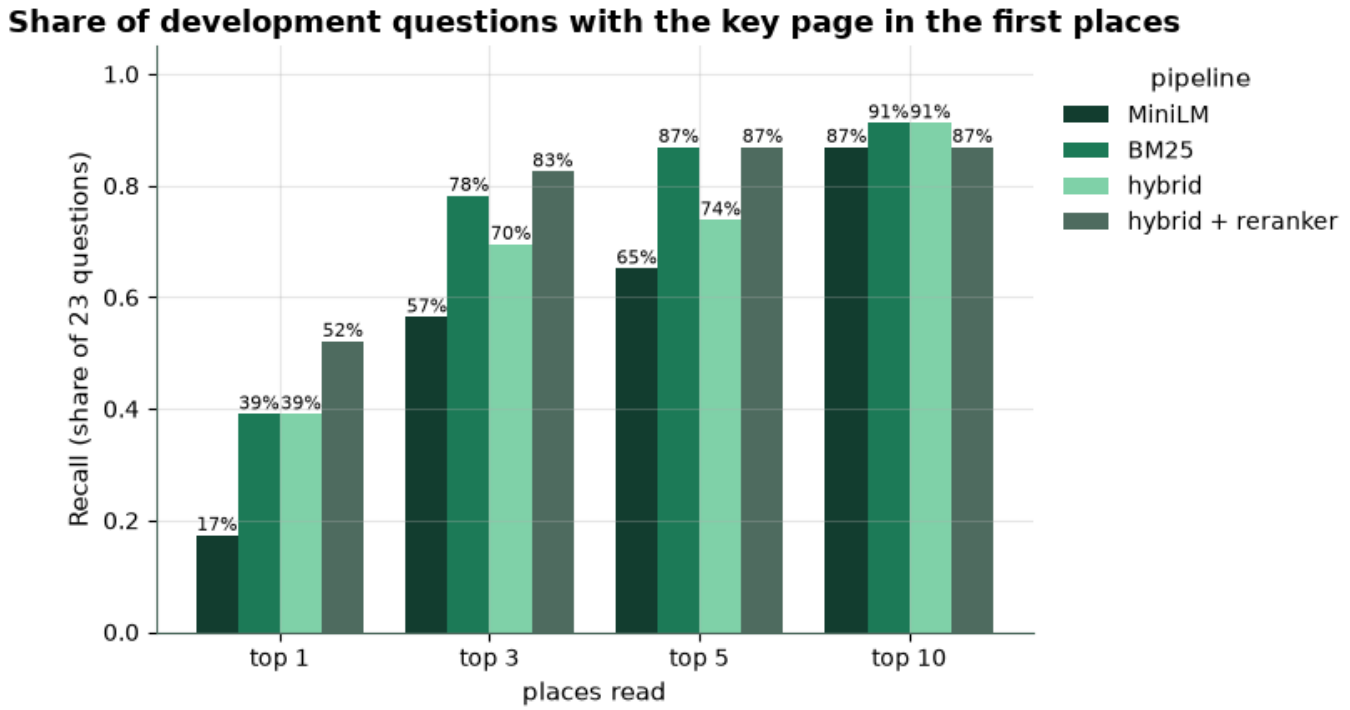
```
recall at 3 by hand: 1 of 3 questions hit = 0.3333

MRR by hand: (1/1 + 1/5 + 0) / 3 = 0.4000

recall_at_k at 3: 0.3333 (by hand 0.3333); mrr: 0.4000 (by hand 0.4000); equal: yes
```

The miss is a 0 in both numerators and a 1 in both denominators. Everything else today is this arithmetic on 23 rows.

Four pipelines on the development questions



Recall at 1, 3, 5, 10 and MRR

MiniLM

`0.17  0.57  0.65  0.87  MRR 0.40`

BM25

`0.39  0.78  0.87  0.91  MRR 0.61`

hybrid

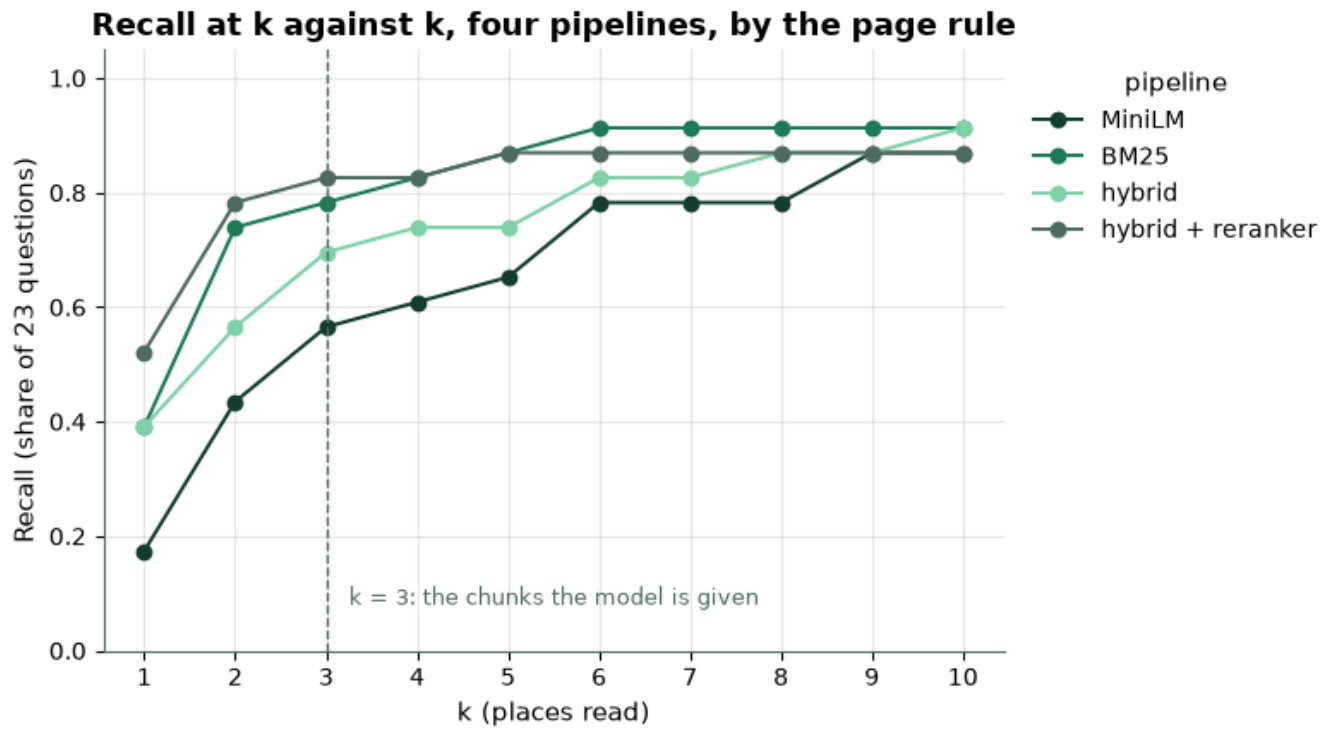
`0.39  0.70  0.74  0.91  MRR 0.56`

hybrid + reranker

`0.52  0.83  0.87  0.87  MRR 0.68`

By the page rule. A miss counts 0 and stays in the mean. 23 questions: one question is 0.04.

Recall at k against k, place by place



How to read the curve

At the dashed line: k = 3 is what the model is given

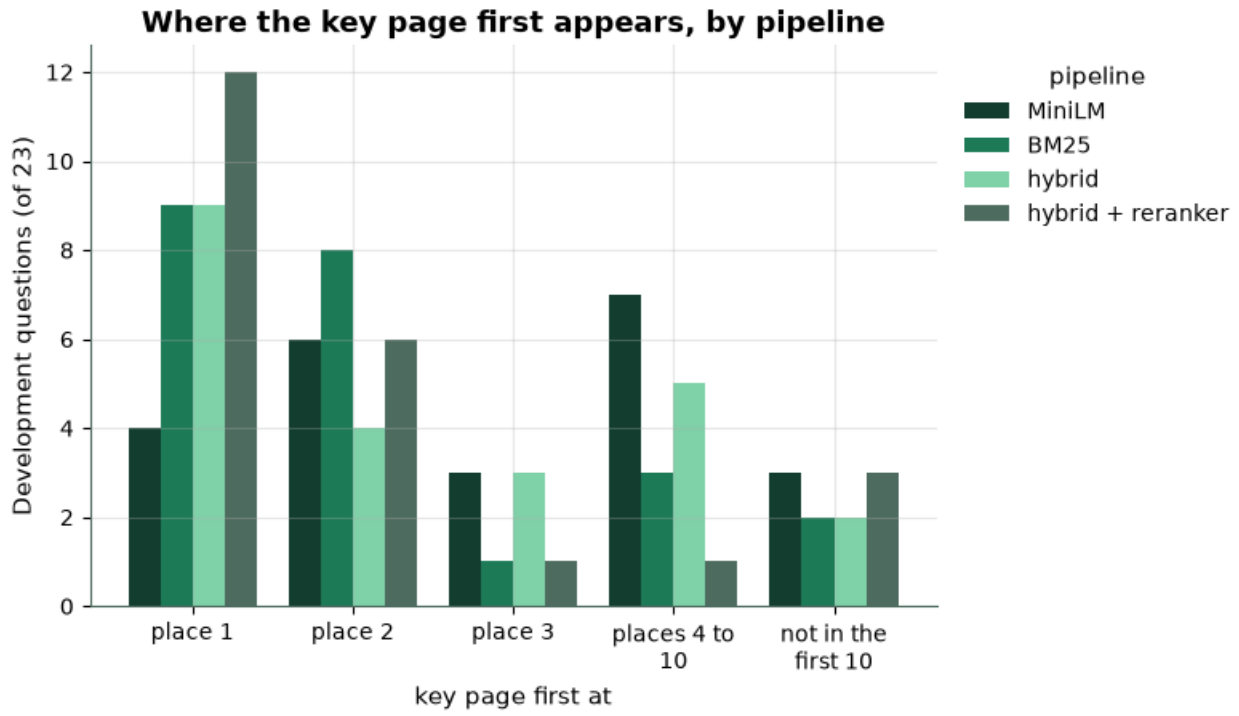
Still rising at the k you send: a larger k or a reranker would add hits

Flat: the misses left are not in the next places; another retriever, not more places

Last place that adds a hit: MiniLM k = 9, BM25 k = 6, hybrid k = 10, hybrid + reranker k = 5

The reranker's curve starts highest; the hybrid's is still rising at 10, which is why it has one more hit there.

Where the key page first appears

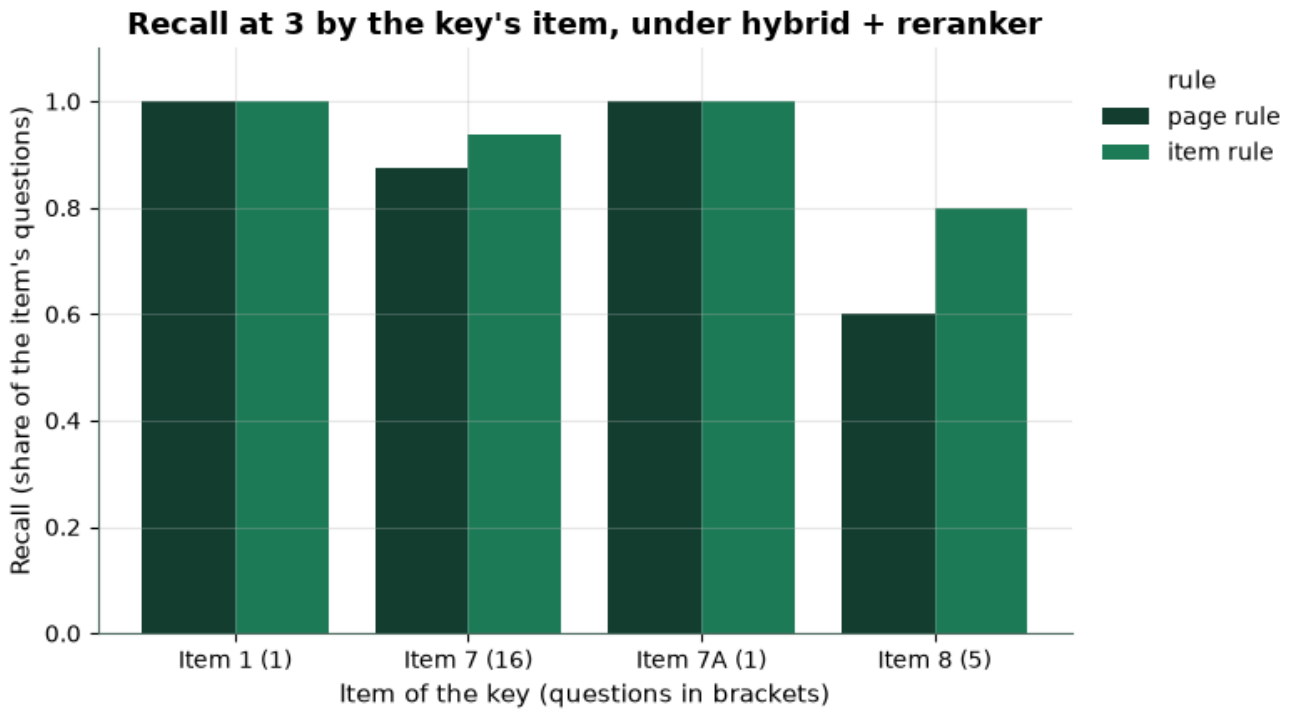


Questions, by first place

	1	2	3	4-10	none
MiniLM	4	6	3	7	3
BM25	9	8	1	3	2
hybrid	9	4	3	5	2
hybrid + reranker	12	6	1	1	3

Recall at 10 says the page is somewhere in ten places; MRR says how early.

The page rule and the item rule



At 3: page rule 19 of 23, item rule 21

item	n	page	item
1	1	1	1
7	16	14	15
7A	1	1	1
8	5	3	4

The item rule counts any chunk from the key's item. Groups this small describe; they do not estimate.

The four misses at 3

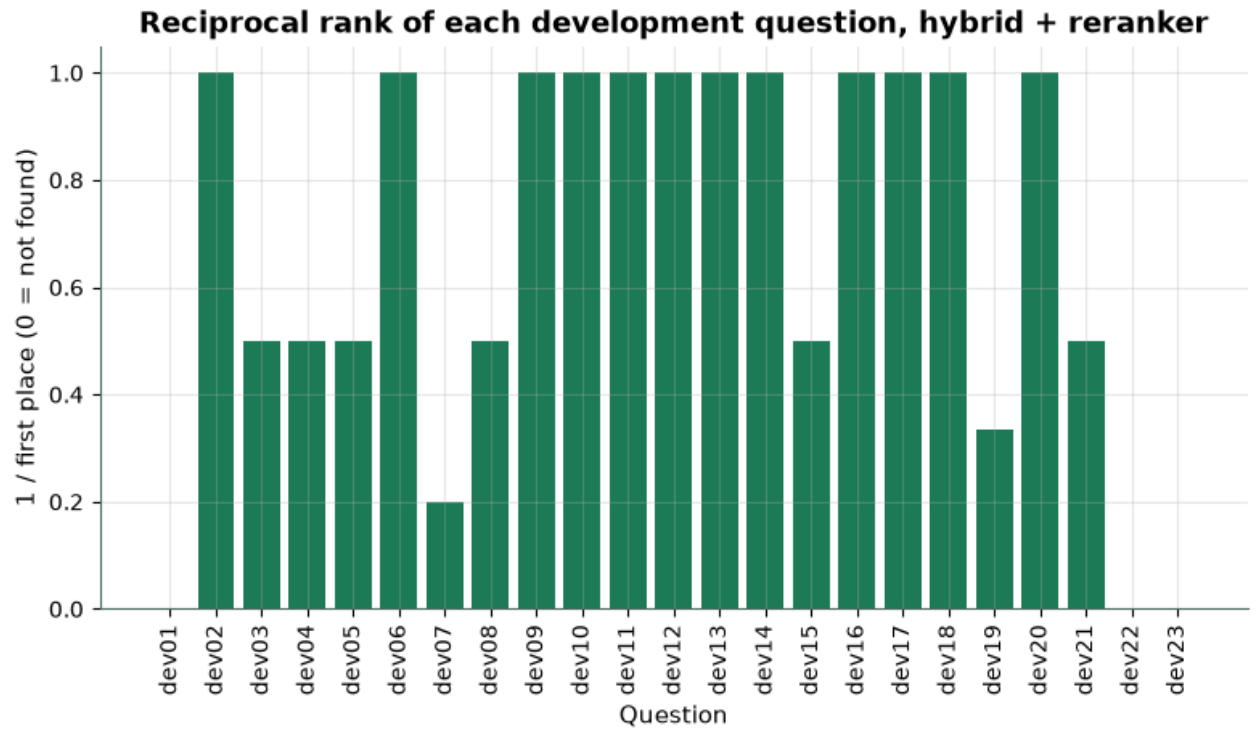
Place of the first chunk on the key page (- = not in the first 10)

id	item	key page	by page	by item	pages of the first 3
dev01	7	37	-	1	40 78 36
dev07	7	38	5	5	76 101 76
dev22	8	52	-	2	43 61 94
dev23	8	98	-	4	34 34 34

dev22: total assets, key on page 52 (Item 8)

The first place is an Item 7 paragraph that uses the question's words and holds no balance sheet figure; the chunk with the key is the balance sheet. Share of lines that look like table rows: first place 0.25, chunk with the key 0.70. The notebook does not test why the cross-encoder scored the paragraph above the table.

Reciprocal rank, with the misses at 0



Two means of the same run

0.6754

MRR over all 23 questions, misses counting 0

0.7767

mean over only the 20 questions that were found

3

misses at 0: dev01, dev22, dev23

A mean over the questions that were found describes the retriever where it is already good.

Opened once, for the setting chosen before

hybrid + reranker, by the page rule

rows	recall 1	recall 3	recall 5	recall 10	MRR
development (23)	0.52	0.83	0.87	0.87	0.68
test (18)	0.39	0.89	0.94	0.94	0.62

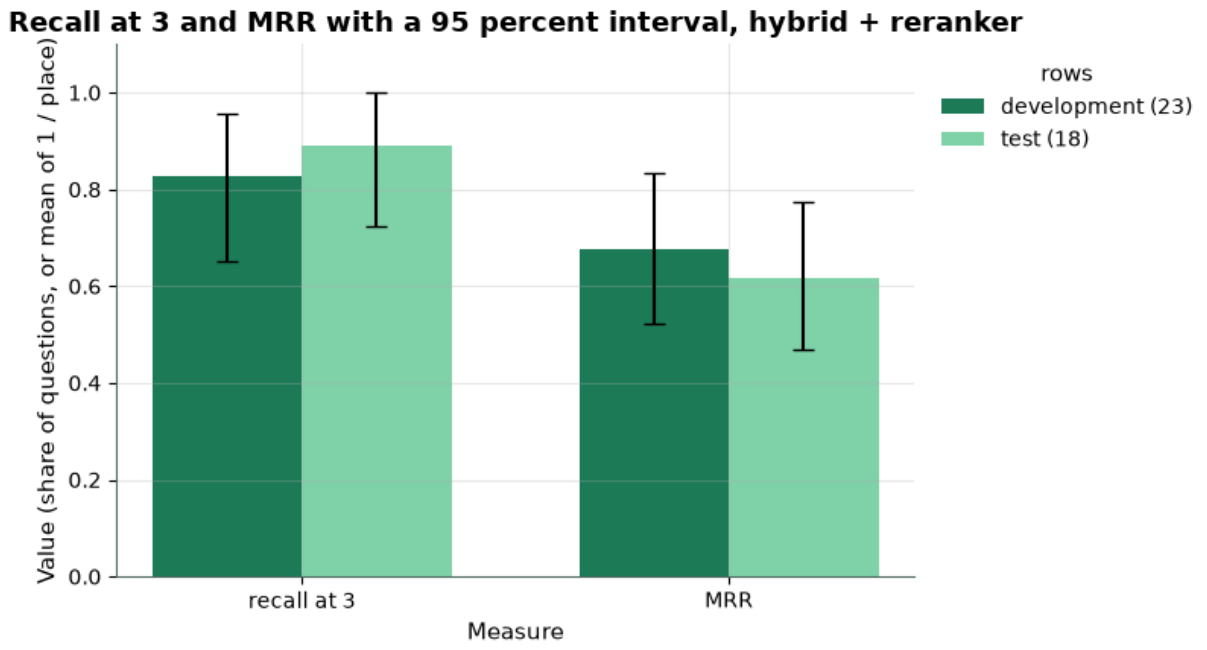
The one cell that reads the test rows

recall at 1 7 of 18 (0.3889), recall at 3 16 of 18 (0.8889), recall at 5 17 of 18 (0.9444), recall at 10 17 of 18 (0.9444); MRR 0.6157. Not in the first 3: test01, test08. If the number disappoints, the setting stays and the number is reported.

How it works: a bootstrap interval

One count	A recall on 18 questions is one count of 18; another 18 questions would have given another count.
Resample	18 drawn from 18 with replacement (a question twice, another not at all); the recall of that draw; 1,000 times, seeded.
Read	The 2.5th and 97.5th percentiles of the 1,000 recalls: the range the number would wander over if the set were drawn again.
ensemblekit	bootstrap_interval does this for any metric of labels and values; written for a ROC AUC, it redraws a one-class resample.
What comes out	A low and a high to say beside the point estimate; today a plain resampling shows what the redraw rule changes.

An interval, and which function made it



ensemblekit and plain resampling

development, recall at 3 0.83

```
ens 0.65-0.96   plain 0.65-0.96
```

development, MRR 0.68

```
ens 0.52-0.83   plain 0.52-0.83
```

test, recall at 3 0.89

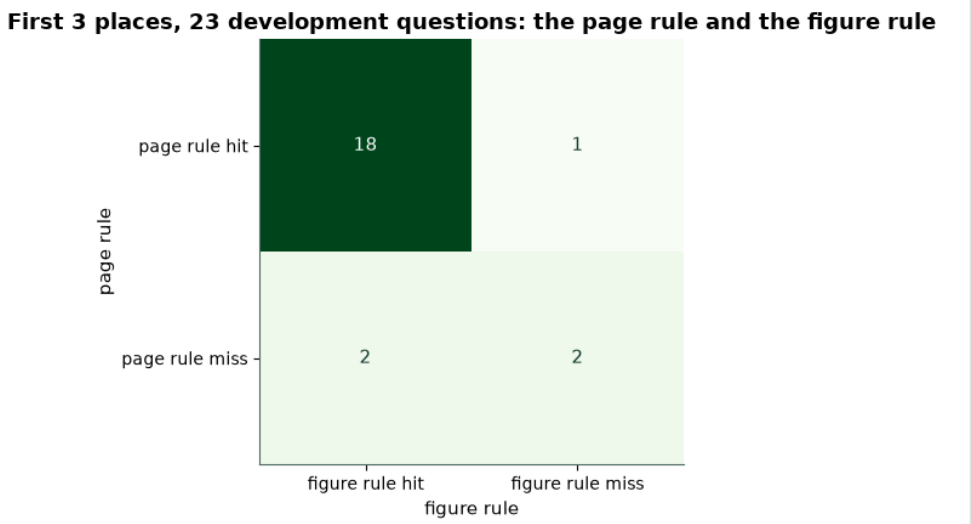
```
ens 0.72-0.94   plain 0.72-1.00
```

test, MRR 0.62

```
ens 0.46-0.76   plain 0.47-0.77
```

ensemblekit redraws a resample with one class only; a resample of 18 with no miss at 3 has chance 0.120. Say which function made the interval.

The page rule and the figure rule



First 3 places, 23 development questions

page rule and figure rule both hit 18; page rule only 1 (dev19); figure rule only 2 (dev01, dev07); neither 2 (dev22, dev23)

dev19: the page is returned, the figure sits in the neighbour chunk

dev01 and dev07: the figure is in the chunks, from another page

A page rule undercounts where a figure has two homes

Recall stops at the model's door

Neither rule is wrong. Each measures something else, and a count must say which.

Excel side by side, and Git

Excel side by side (tested in Excel)

`=IFERROR(MATCH($B2,$C2:$L2,0),0)`	first place of the key page; equal to evaluate's first_rank for 23 rows
`=ISNUMBER(MATCH($B2,$C2:$E2,0))`	hit at 3, equal to Python for 23 rows
`=COUNTIF(O2:O24,TRUE)/ROWS(O2:O24)`	recall at 3: 0.8261, equal to Python
`=SUMPRODUCT((N2:N24>0)*(N2:N24<=5))/ROWS(N2:N24)`	recall at 5: 0.8696, equal to Python
`=IFERROR(1/MATCH($B2,$C2:$L2,0),0)`	reciprocal rank; the miss becomes 0, equal to Python for 23 rows
`=AVERAGE(P2:P24)`	MRR: 0.6754, equal to Python
`=AVERAGEIF(P2:P24,">0")`	leaves the misses out: 0.7767, the buggy function's number
`=COUNTIFS($M$2:$M$24,"8",$O$2:$O$24,TRUE)/COUNTIF($M$2:$M$24,"8")`	Item 8 recall at 3: 0.6000, equal to Python
`=PERCENTILE.INC(A1:A1000,0.025)`	2.5th and 97.5th percentiles of 1,000 means: 0.7222 and 1.0000
`=ISNUMBER(MATCH(T1,$C3:$E3,0))`	FALSE when the key page is stored as text, TRUE as a number

Git: retrieval_eval.csv, 23 development rows, then 18 test rows; 53 tests.

The AI check

The docstring is the prompt. The retrieval code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the filter first_rank > 0 drops every question that was not found, so the mean ignores the misses.

Code that runs is not the same as code that is right.

What you can do now

- Two families of measures: retrieval reads places, answer measures read text
- evaluate, recall at k, MRR; a miss counts 0; a recall curve against k
- Read by item; name the rule of every count
- Open the test rows once; say which function made the interval
- A page-rule hit is not a figure in the chunks, and not a right answer

Week 3, Day 2 LLM judges and their limits